

Machine Learning for Bioinformatics

Exercise Sheet

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Please note that the jupyter notebook must be submitted
along with the exercise sheet!¹

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Artificial Neural Networks

The PyTorch introduction already covered tensors, data loaders, modules, losses, and basic training loops. Here we focus on neural-network-specific modeling choices: hidden layers, activation functions, model size, overfitting, early stopping, and regularization.

Assignment 1. *One-hidden-layer network* We use the SMS spam dataset represented as binary word-occurrence vectors. The preprocessing and training utilities are provided in the notebook.

1. Complete the implementation of the *MLPSpamNetwork* class. Points: 0
2. How many trainable parameters does the model have for a hidden layer with 64 units? Points: 1

Parameters:

Assignment 2. *Training and activation functions* Train the one-hidden-layer model and compare different hidden-layer activation functions.

1. Report the best validation epoch and the test accuracy of the MLP with ReLU activation. Points: 1

Best epoch:

Test accuracy:

2. Which hidden-layer activation function gives the best validation accuracy? Report its test accuracy. Points: 1

Activation:

Test accuracy:

Assignment 3. *Overfitting and regularization* Compare small and large hidden layers, then train the large model with dropout, weight decay, and early stopping.

1. For the model with 256 hidden units, report the best validation epoch and the final train-validation accuracy gap. Points: 1

¹No points are awarded if an answer is only partially correct. Answers must be supported by results from the jupyter notebook.

Best epoch: Gap:

2. Report the test accuracy of the regularized large model. Points: 1

Test accuracy:

3. Specify the values of the regularized model confusion matrix. Points: 1

Assignment 4. Theory questions Answer the following questions based on the artificial neural networks lecture.

1. A neural network with no hidden layer, one output node, and sigmoid output activation is equivalent to a logistic regression model. Points: 1

Correct: Incorrect:

2. Assume that every activation function in a network is linear. Stacking several such layers can still represent a nonlinear function of the input. Points: 1

Correct: Incorrect:

3. Which output activation is appropriate for a binary classification problem with one output node? Points: 1

Identity: Sigmoid: Softmax:

4. Which statement best describes why ReLU is often preferred over sigmoid or tanh in hidden layers of deep networks? Points: 1

ReLU is less prone to vanishing gradients.

ReLU always gives nonzero gradients for all inputs.

ReLU is used only in output layers for binary classification.

5. Assume a neural network has training accuracy 0.98 but test accuracy 0.60. Which explanation is most plausible? Points: 1

The model is likely overfitting.

The model is too simple.

The test set is irrelevant for model assessment.

6. Increasing the hidden layer size from 16 to 256 increases the number of trainable parameters. Which consequence is most likely? Points: 1

The model capacity increases, and the risk of overfitting can increase.

The model becomes linear.

The sigmoid output is no longer valid for binary classification.

Maximum number of points: 12